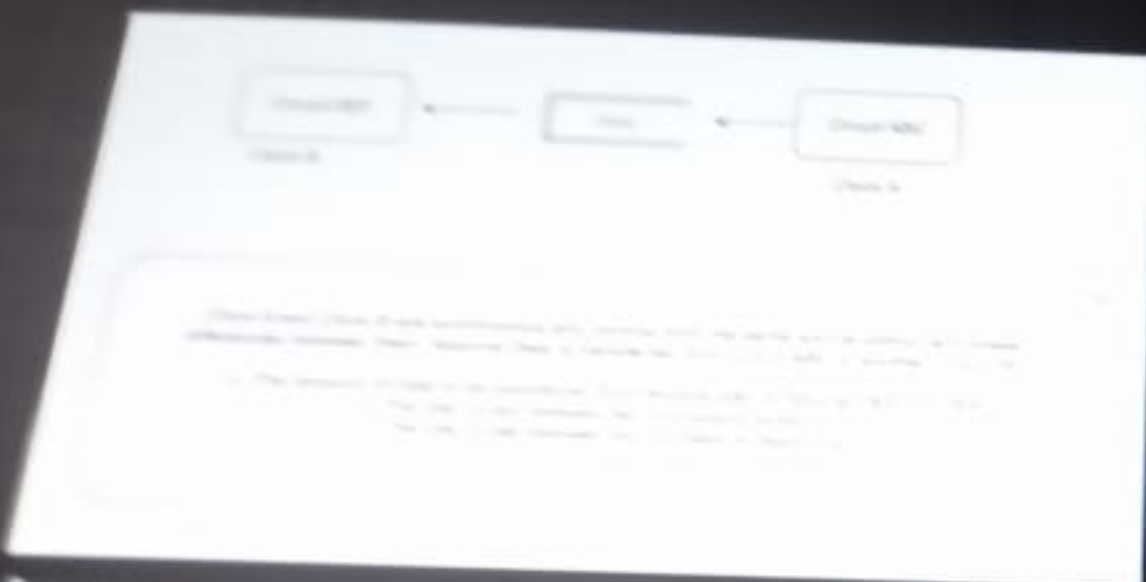




Question 15

Figure indicates the Data flow between the 2 circuits.



Assume

Frequency of clock A (MHz) = 60

Frequency of clock B (MHz) = 100

$D = 39$

$V = 1$

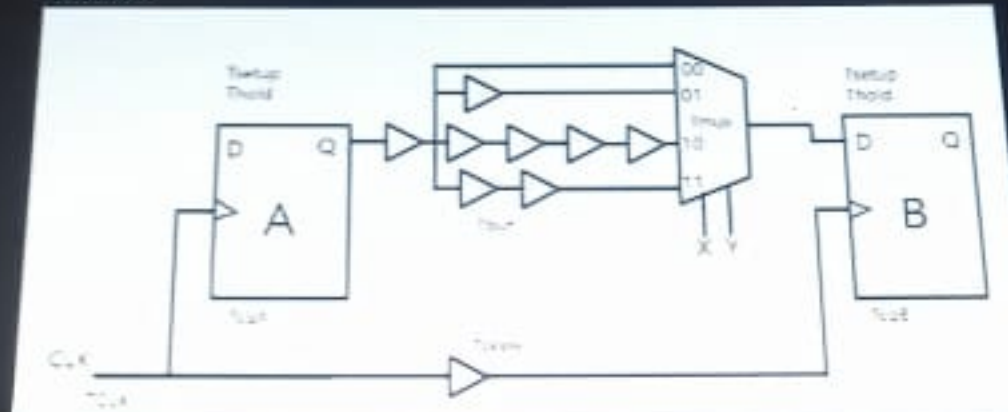
$W = 9$

Complete the blanks in the following question with the appropriate answer.

Then, Minimum Fifo Depth (Rounded off) = _____

Question 18

Analyze the below circuit for timing parameters listed in table below:



Note: 'Tbuf' represents the delay of each buffer.

TIMING PARAMETER	VALUE (ps)
Tbuf	3
TcqA	15
TcqB	25
TCLK	40
Tskew	10
Tmux	4
Tsetup	26
Thold	12

Complete the blanks in the following question

Value of mux selects X & Y for which the circuit ca

XY= _____

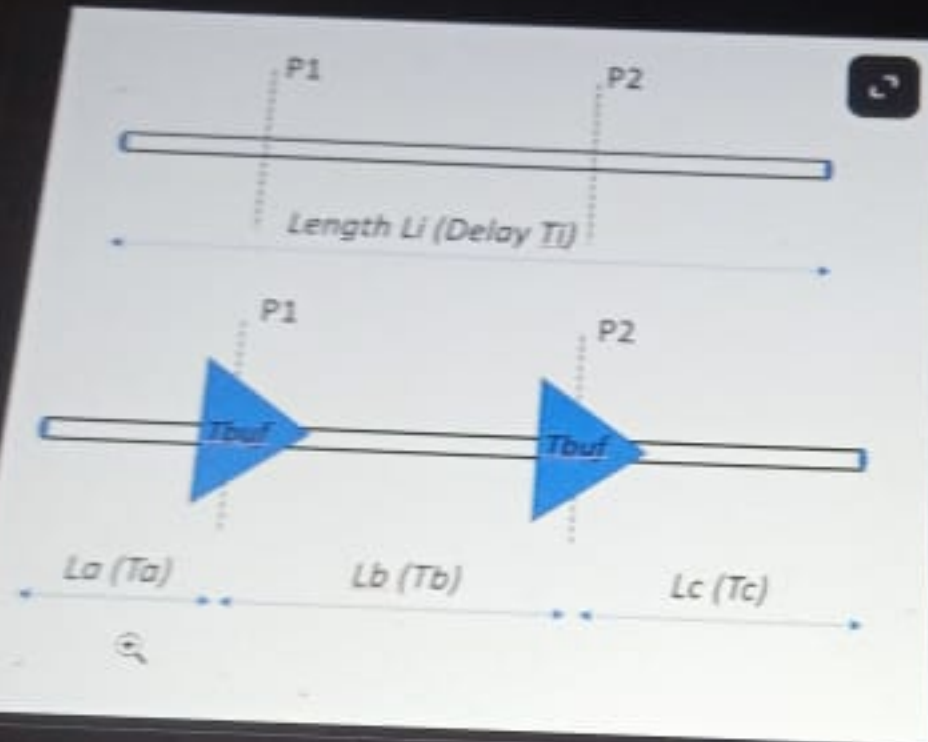
For such reliable timing operation,

Minimum Combinational Delay (ps) between F

Maximum Combinational Delay (ps) between

Question 13

A wire of length L_i has delay of T_i . A Buffer of delay T_{buf} is inserted at each of the points P1 and P2 thereby creating 3 wire segments of Lengths L_a , L_b & L_c with Delays T_a , T_b & T_c respectively. The new circuit is shown below:



L_i	100 μm
T_i	100 ps
P1	20 μm
P2	80 μm
T_{buf}	25 ps

Complete the blanks in the following question with the appropriate answers.

Estimated Wire Segment Delays:

T_a is _____ ps

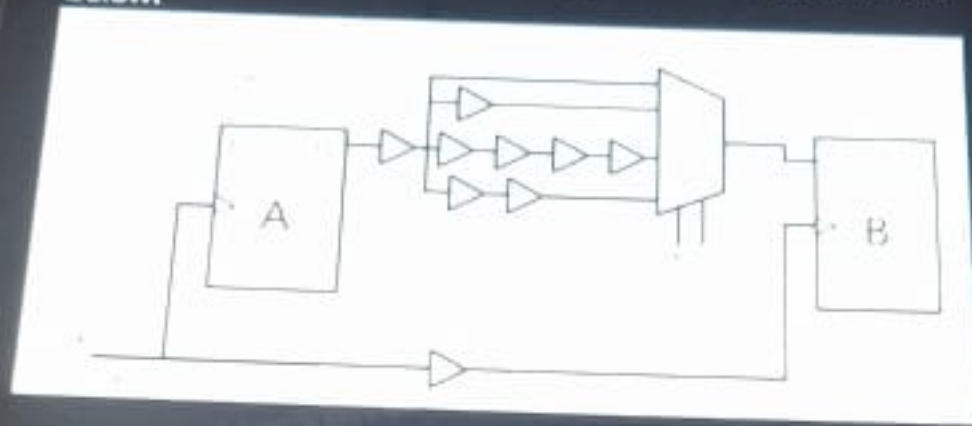
T_b is _____ ps

T_c is _____ ps

Estimated Total Delay of the new circuit (wires + buffers included) is _____

Question 18

Analyze the below circuit for timing parameters listed in table below:



Note: T_{buf} represents the delay of each buffer.

TIMING PARAMETER	VALUE (ps)
T_{buf}	3
T_{cqA}	15
T_{cqB}	25
TCLK	40
T_{skew}	10
T_{mux}	4
T_{setup}	26
T_{hold}	12

Complete the blanks in the following question with the

Value of mux selects X & Y for which the circuit can function reliably

XY= _____

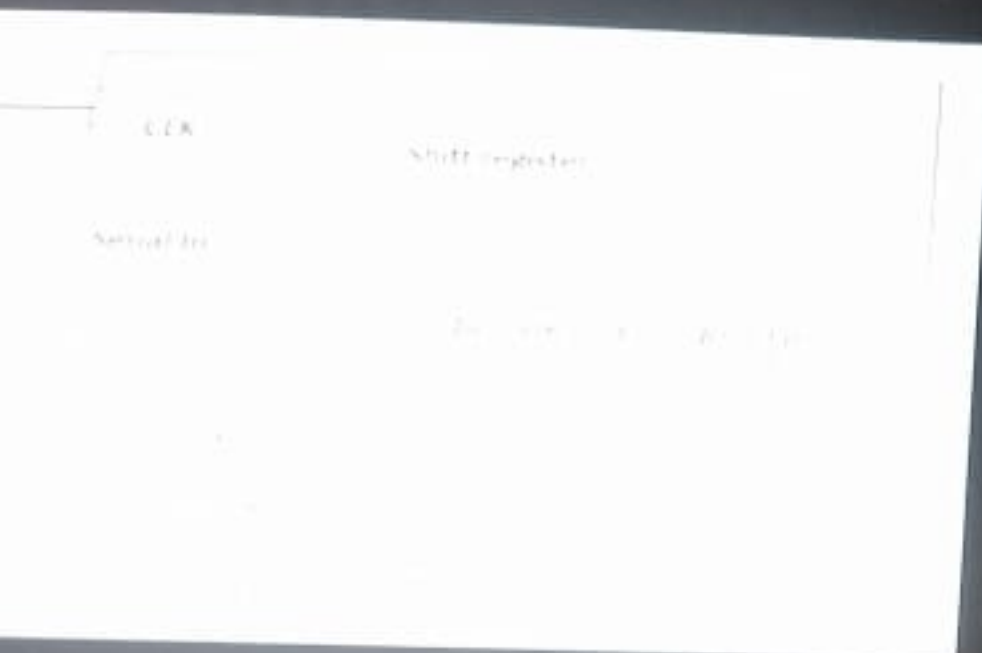
For such reliable timing operation,

Minimum Combinational Delay (ps) between Flop A & B is _____

Maximum Combinational Delay (ps) between Flop A & B is _____

Question 16

A 5-bit serial-in-parallel-out shift register is used with feedback as shown in Figure. The shifting sequence is QA → QB → QC → QD → QE.



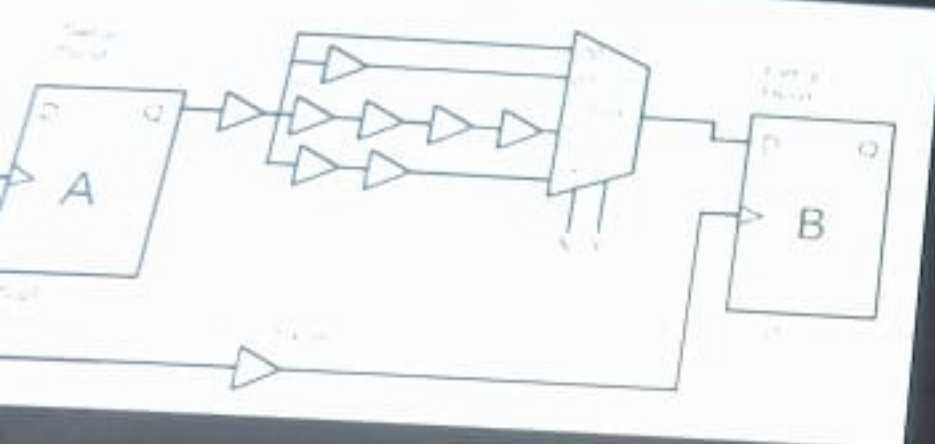
Value of QA, QB, QC, QD, QE is 10101

Complete the blanks in the following question with the appropriate answer.

Then final value of QA, QB, QC, QD, QE after 8 clock cycles is _____.

Question 18

Use the below circuit for timing parameters listed in table



represents the delay of each buffer.

METER	VALUE (ps)
	3
	15
	25
	40
	10
	4
	26
	12

Complete the blanks in the following question with the appropriate answer.

Value of mux selects X & Y for which the circuit can function reliably in terms of timing is
 $XY = \underline{\hspace{2cm}}$

For such reliable timing operation,

Minimum Combinational Delay (ps) between Flop A & B is

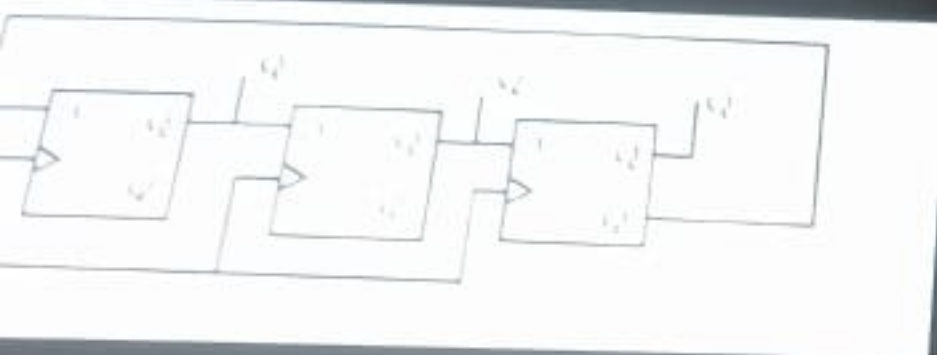
Maximum Combinational Delay (ps) between Flop A & B is

Question 12

Consider the below logic circuit built using D flipflops.

Initial value of (Q_2, Q_1, Q_0) at the 0th cycle is $\{0, 0, 0\}$

Enter the 3 bits without any comma, for example if
1,1,0: enter 110



11

Complete the blanks in the following question with the appropriate answer.

The value of (Q_2, Q_1, Q_0) after Nth cycle is _____

Question 14

```
#include <stdio.h>

int compute(int i) {
    if(i <= 1) {
        return 1;
    }
    return i + compute(i - 1);
}

int main() {
    printf("%d\n", compute(i)); // Value of i
    given below
    return 0;
}
```

Given i=9

Complete the blanks in the following question with the

Output of the program is _____



S10

10

S11

11

S12

12

S13

13

S14

14

S15

15

S16

16

S17

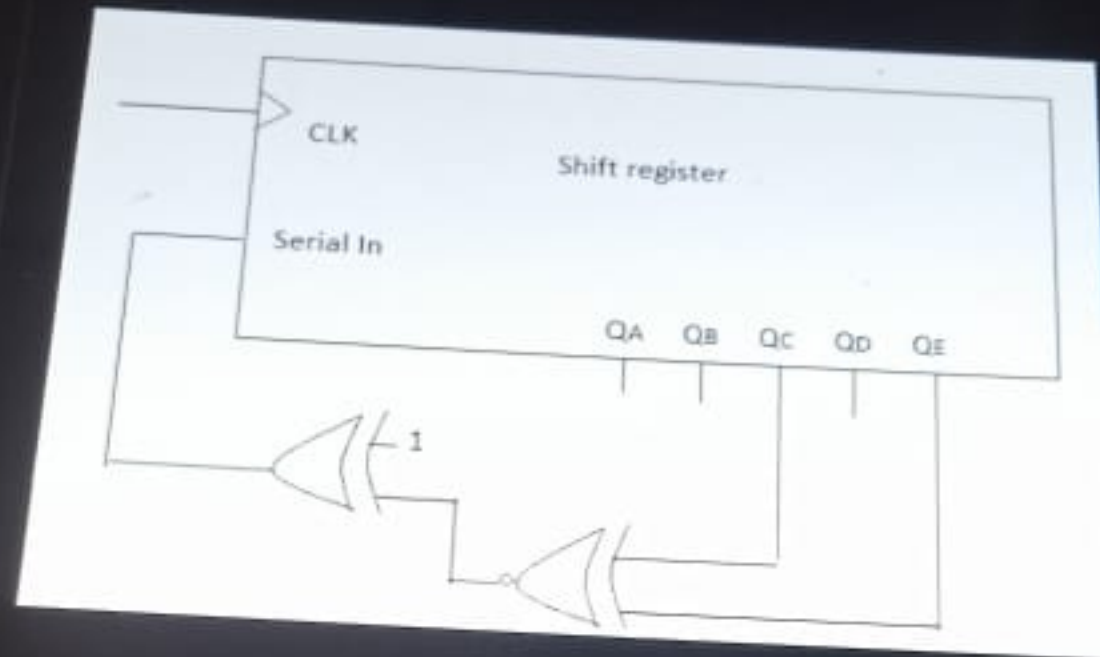
17

S18

18

Question 16

A 5-bit serial-in-parallel-out shift register is used with feedback as shown in Figure. The shifting sequence is QA → QB → QC → QD → QE



If the initial value of QA, QB, QC, QD, QE is 10101

Complete the blanks in the following question

Then final value of QA, QB, QC, QD, QE after 8 clock

S10

10

S11

11

S12

12

S13

13

S14

14

S15

15

S16

16

Question 16

S17

17

S18

18

Question 10

Write a method:

`int compute(int n)`

that returns:

the value of n

if n is even

the value of n

plus the value of `compute(n-1)` if n is odd

Write the

method

Class:

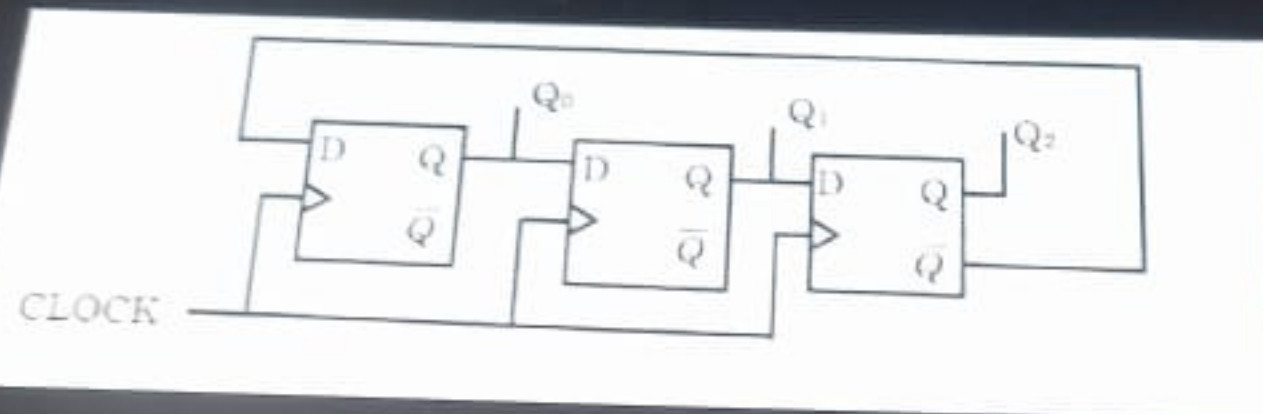
Complete the method in the following way

Output of the program is _____

Question 12

Consider below logic circuit built using D flipflops.
The initial value of (Q_2, Q_1, Q_0) at the 0th cycle is $\{0, 0, 0\}$

Note: Enter the 3 bits without any comma, for example if answer is 1,1,0; enter 110



N

110

Complete the blanks in the following

The value of (Q_2, Q_1, Q_0) after Nth cycle is

Question 17

Following circuit of identical NMOS transistors shown in the figure below:
 the threshold voltage of all the transistors.



1V
200 mV
900 mV

Complete the blanks in the following question with the appropriate answer.

What will be the voltage at node "N1" ? _____ mV

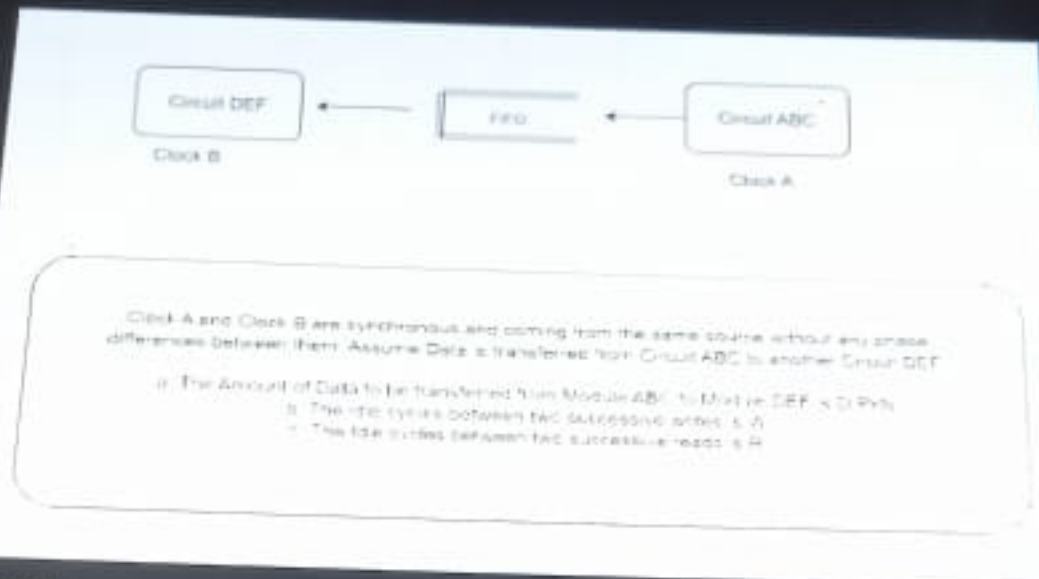
What will be the voltage at node "N2" ? _____ mV

What will be the voltage at node "N3" ? _____ mV

What is the region of operation of transistor T2 (linear or saturation or cutoff ?) _____ region

What is the region of operation of transistor T3 (linear or saturation or cutoff ?) _____ region

Figure indicates the Data flow between the 2 circuits.



Assume

Frequency of clock A (MHz) = 60

Frequency of clock B (MHz) = 100

$D = 39$

$W = 1$

$R = 9$

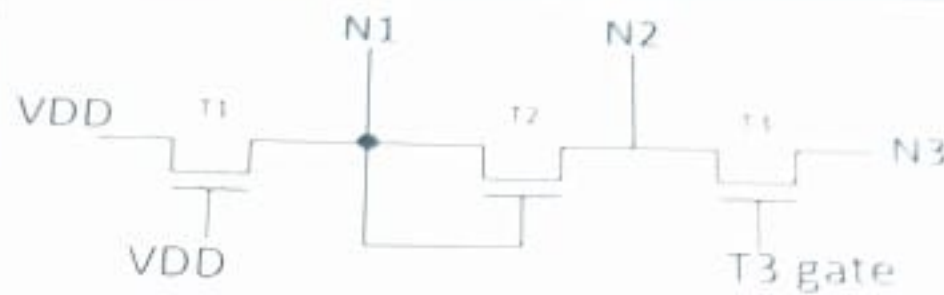
Complete the blanks in the following question

Then, Minimum Fifo Depth (Rounded off) = _____

Question 15

Question 17

Consider the following circuit of identical NMOS transistors connected as shown in the figure below:
 V_{TH} represents the threshold voltage of all the transistors.



VDD	1V
V_{TH}	200 mV
T3 gate	900 mV

Complete the blanks in the following:

What will be the voltage at node "N1"?

What will be the voltage at node "N2"?

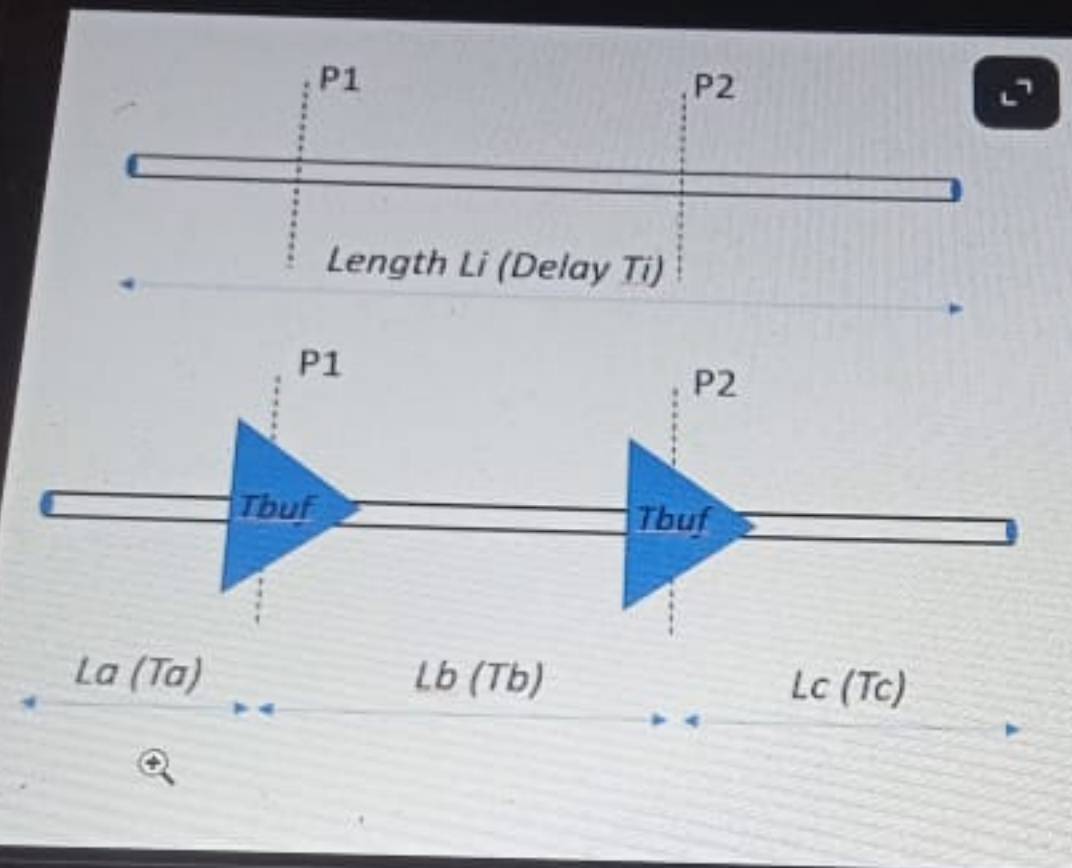
What will be the voltage at node "N3"?

What is the region of operation of T1?

What is the region of operation of T2?

Question 13

A wire of length L_i has delay of T_i . A Buffer of delay T_{buf} is inserted at each of the points P1 and P2 thereby creating 3 wire segments of Lengths L_a , L_b & L_c with Delays T_a , T_b & T_c respectively. The new circuit is shown below:



L_i	100 μm
T_i	100 ps
P1	20 μm
P2	80 μm
T_{buf}	25 ps

Complete the blanks in the following question with

Estimated Wire Segment Delays:

T_a is _____ ps

T_b is _____ ps

T_c is _____ ps

Estimated Total Delay of the new circuit (wires + buffers) is _____ ps